

SESSION 13

Foundations of Probability

Sample Spaces · Events · Classical & Axiomatic Probability · Addition & Multiplication Rules

Random Experiments & Sample Spaces

Defining Events

Three Approaches to Probability

Kolmogorov Axioms of Probability

The Addition Rule (Union)

Independent vs Mutually Exclusive Events

The Multiplication Rule (Intersection)

Python Probabilistic Simulations

Session Notes — Complete Reference

Playlist: Maths for Machine Learning · Instructor: Nitish Singh (CampusX)



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01 • Random Experiments & Sample Spaces

A **random experiment** is a process with an uncertain outcome. The set of all possible outcomes is the **Sample Space** (S).

Example: Rolling a six-sided die has $S = \{1, 2, 3, 4, 5, 6\}$.

02 • Defining Events

An **event** (A) is any subset of the sample space S . We say event A occurred if the experiment's outcome belongs to A .

Example: Event $A = \text{"rolling an even number"} = \{2, 4, 6\}$.

03 · Three Approaches to Probability

- **Classical:** $P(A) = \frac{\text{number of outcomes in } A}{\text{total outcomes in } S}$ (assumes equally likely outcomes).
- **Empirical (Relative Frequency):** $P(A) = \lim_{N \rightarrow \infty} (n_A / N)$ as we repeat the trial N times.
- **Subjective:** Personal belief or opinion about the likelihood of an event.

04 • Kolmogorov Axioms of Probability

For any probability function P :

1. $P(A) \geq 0$ for all events A .
2. $P(S) = 1$.
3. If $A_1, A_2, \dots$ are mutually exclusive, then $P(\bigcup A_i) = \sum P(A_i)$.

05 • The Addition Rule (Union)

The probability that event \$A\$ OR event \$B\$ occurs:

ADDITION RULE FORMULA

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

06 • Independent vs Mutually Exclusive Events

Mutually Exclusive

Events cannot happen at the same time.

$$P(A \cap B) = 0$$

Independent

Occurrence of one event does not affect the probability of the other.

$$P(A \cap B) = P(A) \cdot P(B)$$

07 · The Multiplication Rule (Intersection)

The probability that event \$A\$ AND event \$B\$ both occur:

MULTIPLICATION RULE FORMULA

$$P(A \cap B) = P(A) \cdot P(B \mid A)$$

08 · Python Probabilistic Simulations

```
import random

# Monte Carlo simulation of flipping two heads in a row
flips = 100000
two_heads = 0

for _ in range(flips):
    f1 = random.choice(['H', 'T'])
    f2 = random.choice(['H', 'T'])
    if f1 == 'H' and f2 == 'H':
        two_heads += 1

print(f"Simulated Probability: {two_heads / flips:.4f}")
print(f"Theoretical Probability: {0.25}")
```

09 · Quick Reference Card



Probability part 1 Cheat Sheet

- **Union (OR):** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- **Intersection (AND):** $P(A \cap B) = P(A) \times P(B|A)$.
- **Complement:** $P(A^c) = 1 - P(A)$.
- **Mutual Exclusion:** $P(A \cap B) = 0$.